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	Human performance is easily impaired in a variety of aerospace tasks – such as remotely docking a spacecraft, or flying an aircraft in poor visibility, or finding and tracking a particular target airplane on a crowded air-traffic control display. These impairments occur because aerospace environments can degrade or remove the sensory cues used to drive and coordinate voluntary actions, and display systems may not be optimally matched to the human user. Fortunately, much progress has been made recently in the field of neuroscience toward understanding the physiological mechanisms responsible for voluntary motor behaviors. In particular, we are beginning to understand how the brain accomplishes visual search – the process that allows us to quickly respond to important items in our surroundings, even in the presence of other irrelevant stimuli. Understanding how individual neurons participate in this process is critical for identifying the factors that limit the speed and accuracy of visual search. Knowledge of these neural factors could be used to design visual displays that conform to the temporal and spatial constraints imposed by human neurobiology, thereby directly		

improving human performance in man-machine interfaces and virtual environments.
Arguably the most important class of movements made by primates during visual search are eye movements, because control of the position of the eyes determines which visual information is sampled and for how long. Decades of anatomical and physiological research have provided a detailed outline of the brain pathways that are responsible for the control of eye movements in primates. These pathways involve a network of areas including visual areas of the cerebral cortex (which provide information about the content of the visual scene), motor regions of the cerebellum and brainstem (which form the final motor commands), and several regions that lie in-between (and which possess a mixture of sensory and motor properties). It is these latter, intermediate, brain regions that are most likely responsible for forming the decision about where the eyes should move next during visual search, because they lie at the interface between the visual areas that provide evidence about the location of stimuli and the motor regions that execute the movements. One of these structures, the superior colliculus (SC), has long been known to play a pivotal role in the control of saccadic eye movements. However, most studies of the SC have focused on its contribution to the motor execution of saccades. Because of the importance of the SC in the decision processes that precede saccade generation, examination of this structure during visual search is likely to reveal fundamental aspects of the visual neural computations involved in identifying, localizing, and selecting visual targets. The main goal of this project would be to measure how neurons in the superior colliculus (SC) change their activity during a visual search task. Specifically, we propose to measure how the activity of these neurons is altered by the discriminability of visual targets and test whether these changes can predict the changes in the subjects' performance. Understanding how the information encoded by these neurons constrains overall search performance will foster the development of more predictive models of human performance and ultimately lead to the design of better aerospace displays and interfaces.

Task Description:

Goal: To develop a computational model of decision-making, involving multiple sources, agents and outputs, based on neurophysiological data.

The phenomenon: Visual search is a basic form of decision-making that is performed by humans and non-human primates several times a second (e.g., as you read this text). The brain pathways that accomplish this behavior have been identified, but the processing steps and computations that are involved in supporting this phenomenon are not known.

The experimental approach: By recording from single neurons in the brains of non-human primates as they perform visual search, we have a window into the internal processing steps that support visual search and how these change over time on a millisecond time scale – data that cannot be obtained from behavioral experiments with humans. In addition, we conduct parallel behavioral experiments with humans to confirm the applicability of this approach.

The hypothesis: By recording the activity of neurons during visual search, we can identify the biological computations that underlie performance in visual-motor tasks and decision-making.

Countermeasure readiness level: 2 at start of project, 3 now. The original hypothesis has now been validated and we are in the process of understanding in detail the neural processes steps that support neurally-mediated visual search. Specifically, we are testing how different sources of information are combined and read out by neurons during this basic decision task.

Expected countermeasure readiness level at end of project: 4 to 5. Once we understand the dynamic regulation of top-down and bottom-up factors, we will have identified the key features that are relevant for constructing biologically inspired models of visual search. We will then formulate how these features should be incorporated into computer-based models of decision-making (level 4), and then prove the feasibility and efficacy of this approach by constructing a model of visual search based on these biological features (level 5). After this initial demonstration, the concept may be applied to other processes that would benefit from automated decision-making (levels 5 – 6).

The results from these experiments provide new insights into the biological mechanisms underlying visual search behavior in humans and other primates. Although the immediate goals of the project were to expand our basic biological knowledge, the results are also be applicable to the stated goals of NASA's Space Human Factors, Intelligent Systems, and Aerospace Operation Systems Programs. Biological systems are capable of accomplishing tasks that have proved extremely difficult to duplicate using artificial intelligence and robotic systems – especially the job of detecting, discriminating and responding to stimuli quickly and accurately, even in the presence of background noise. Our results have provided new data about the algorithms and information processing streams used by biological systems to accomplish such tasks, and may therefore aid in the development of robust computational models of human visuomotor performance and in the design of the autonomous systems for future aerospace applications.

Rationale for HRP Directed Research:

Research Impact/Earth Benefits:

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1) Analysis of brain mechanisms during visual search
The initial goal of this project was to investigate the neural mechanisms underlying visual search. Toward this goal, we analyzed neuronal activity in the SC of two subjects while performing a visual search and discrimination task. In brief, we recorded from saccade-related neurons (n=85) in the SC of two subjects during a luminance discrimination task. Subjects initially fixated a central fixation cross on a random noise background, and two boxes at 6 deg on either side of fixation demarcated possible target locations. On cued trials one box briefly reversed contrast, indicating the likely location of the upcoming target (75% valid). After a random interval, two Gaussian-blurred discs were added, with a luminance increment (SNR = 0 to 4) added to one disk. The subject was rewarded for making a saccade to the brighter disc. Our analysis of the neural data was aimed at addressing a basic question about how prior information affects

Task Progress:	performance during visual search: does it change the sensitivity to the visual input itself or does it weight the visual inputs without changing the underlying evidence? The former explanation is the traditional interpretation, whereas the latter explanation is a more recent interpretation based on signal detection theory and applications of Bayesian inference. The results of our analysis favor the weighted evidence interpretation. The subjects' performance in the task showed a clear cue validity effect: higher percent correct when the cue correctly indicated the target location. Neurons in the superior colliculus showed a neural correlate of this cueing effect: higher activity on validly cued trials compared to invalidly cued trials. However, ROC analysis indicated that this effect was almost entirely due to a bias in activity caused by the presence or absence of the cue in the neuron's response field. The ability of the neuron to visually discriminate between target and distracter was largely unaltered by the presence or absence of a cue. These findings are consistent with the idea that saccadic choice behavior is guided by a weighted integration of sensory evidence, and that the activity we recorded in the superior colliculus lies after the point at which those weights are applied. In addition, we have modeled these results using approaches from signal detection theory and Bayesian statistics. Based on the feedback and comments received during the Bioastronautics Investigator's Workshop in Galveston Texas in January 2005 – in particular, that human studies should be emphasized over animal studies, we adjusted our research goals and initiated an additional set of studies aimed at making our neurophysiological studies more directly relevant to issues of human performance. The remaining projects were aimed at understanding human performance during visual search and the underlying biological mechanisms. 2) Shared decision signals for different components of visual search Eye movements generated during visual search have two components: saccades that rapidly redirects the line of sight to new objects, and smooth pursuit that smoothly changes the line of sight to follow moving objects. Most models of visual search account for saccades, but do not include smooth pursuit. We extended our model of eye movement decisions to incorporate pursuit as well as saccades. The data used in this study included approximately 8,000 combined pursuit and saccade movements during a visual search task. This data set allowed us to answer the question: how can pursuit and saccades by guided by a common decision signal and yet still display differences in timing and choices? On the one hand, our results provide additional evidence that pursuit and saccade choices are determined by a common decision signal – in particular, the speed-accuracy curves for pursuit and saccade have the same shape. On the other hand, we also identified two specific differences between pursuit and saccades in how the putative decision signal is read-out to determine the eye movement choices. First, the pursuit system employs a lower response threshold than saccades; this difference accounts for approximately seventy percent of the time-shift as well as the occurrence of trials in which the two motor choices disagree. Second, the saccadic system is subject to an additional motor delay that is not present in the pathways for pursuit; this difference accounts for the remainder of the time-shift observed between the two responses. 3) Eye movement selection when reward probability is dynamically manipulated We have also developed a novel task to examine how knowledge of target probability affects visual search. In these new studies, we have compared the ability of humans to learn a Markov process. Markov chains (stochastic processes where probabilities are assigned based on the previous outcome) are commonly used to examine the transitions between behavioral states, such as those that occur during foraging or social interactions. However, relatively little is known about how well primates can incorporate knowledge about Markov chains into their behavior. Saccadic eye movements are an example of a simple behavior influenced by information about probability, and thus are good candidates for testing whether subjects can learn Markov chains. In addition, when investigating the influence of probability on saccade target selection, the use of Markov chains could provide an alternative method that avoids confounds present in other task designs. To investigate these possibilities, we evaluated human behavior on a task in which stimulus reward probabilities were assigned using a Markov chain. On each trial, the subject selected one of four identical stimuli by saccade; after selection, feedback indicated the rewarded stimulus. Each session consisted of 200-600 trials, and on some sessions, the reward magnitude varied. On sessions with a uniform reward, subjects (n = 6) learned to select stimuli at a frequency close to reward probability, which is similar to human behavior on matching or probability classification tasks. When informed that a Markov chain assigned reward probabilities, subjects (n = 3) learned to select the greatest reward probability more often, bringing them close to behavior that maximizes reward. On sessions where reward magnitude varied across stimuli, subjects (n = 6) demonstrated preferences for both greater reward probability and greater reward magnitude, resulting in a preference for greater expected value (the product of reward probability and magnitude). These results demonstrate that Markov chains can be used to dynamically assign probabilities that are rapidly exploited by human subjects during saccade target selection 4) Eye movement selection when sensory information is minimal or absent Deciding where to look next demands the integration of bottom-up intrinsic image salience with top-down information of goals, motivation and past experience of reward. Most research has focused on the characteristics of the image regions we choose to fixate; less attention has been given to the role of top-down influences despite research showing that top-down influences are important in determining the next saccade in natural scanning tasks. In these experiments, we explicitly manipulated prior experience by "hiding" reward locations and asking human and monkey subjects to conduct an eye movement search for the rewarded locations. Over the course of single experimental sessions, subjects redirected their gaze to the estimated center of the rewarded target distribution. Later in the session, the distribution of subjects' eye movements reflected the shape and spread of the distribution. Thus, using this novel eye movement search task, we can control the association of a spatial region with reward and assess how implicit learning contributes to where we look next.
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